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SPACE SHUTTLE CHALLENGER CASE STUDY REPORT

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SUMMARY

On 28th January, Space Shuttle Challenger exploded after 73 seconds of launch killing all of its seven crew members. This demonstrated that even a well-set and highly respected scientific organisation like NASA can perform catastrophes succumbing to improper management. In this report, first the stakeholders were identified. Each decision taken by them was questioned ethically and how it affected the overall accident was discussed. Incidents that happened a day before the launch and the aftermath of the explosion were discussed. Ethical dilemmas were investigated and the extent to which NASA and Thiokol engineers adhered to their safety standards was examined. Ethical framework was determined to assess the decisions and some alternative decisions were put forth. The accident seemed to occur because of the lack of attention paid to the fail-safe working of O-rings. This was a classic case of groupthink where seriousness of an issue got severely undermined due to shared responsibility. This case study showed us that even the slightest chance of failure cannot be overlooked. It was seen that whistle-blowing can prevent disasters. Proper communication between the NASA engineers could have prevented this accident from happening.

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1. INTRODUCTION

On January 28, 1986, the space shuttle Challenger exploded 73 seconds into its flight, killing all seven crew. STS-51-L was the twenty-fifth trip of the Space Shuttle fleet and the orbiter's tenth flight. The crew members are Onizuka, McAuliffe, Jarvis, Resnik, Smith, Scobee, Ronald Erwin McNair. Christa McAuliffe was an American astronaut and teacher from Concord, New Hampshire. The goal of this mission was to deploy a communications satellite and conduct research on Halley's Comet while in orbit. The latter resulted in heightened media coverage of the expedition; the launch and subsequent disaster were televised live in multiple schools across the United States. The Rogers Commission was constituted to investigate the reasons of the catastrophe as the country grieved the deaths of the crew members. According to the Commission, the explosion was caused by a seal failure in one of the solid rocket booster connections. NASA and Morton Thiokol were aware of the O-ring seal concerns, according to Roger Boisjoly, Senior Scientist and recognised rocker seal specialist, but opted to launch despite the advice of Boisjoly and other experts. NASA and Morton Thiokol's management practises influenced the decision-making framework, resulting in a lack of communication among technical specialists and their supervisors, as well as top-level management, and the accepted risks that both corporations had previously judged improper. With human lives and national interests at stake, this substantial shift in management practise presents serious ethical concerns. Indeed, one of the most important aspects of the Challenger disaster is its ethical component, which includes both the causal chain that led to it and the lessons that may be derived from it. The complex web of decisions, events, processes, and organisational structures that led to the demise of the Challenger and its crew is entwined with ethical quandaries. As a result, an ethical assessment of this disaster is essential for a thorough understanding of the tragedy and its implications for any activity where public policy, commercial practise, and human acts intersect. [1]

2. TIMELINE OF THE ACCIDENT

- Within the first 1.0 seconds of flight, an O-ring leak caused a discharge.
- At 59 seconds, flames could be seen coming from the O-ring.
- 64 seconds - External fuel tank punctured72 seconds - The turbocharger starts spinning and the fuel tank collapses due to structural problems
- 73 seconds - STS explodes and crashes. The aircraft is currently flying at Mach 1.92 and at an altitude of 46,000 feet (14,000 meters).
- After launch, the Scope Safety Officer activated the self-destruct mechanism of two SRB 110s. (36 seconds after detonation).[2]

3. CAUSES OF ACCIDENT

- The breakdown of two O-ring seals at a juncture in the shuttle's right solid rocket booster triggered the accident (SRB). The record-low temperatures of the launch hardened the rubber O-rings, decreasing their capacity to seal the joints. The seals broke shortly after lift-off, allowing hot pressurised gas from within the SRB to escape through the joint and burn through the aft connection strut linking it to the external propellant tank (ET), and then the tank itself. The breakdown of the internal systems of the external propellant tank, followed by the spin of the SRB, launched the shuttle stack, which was travelling at Mach 1.92, in a direction that permitted air flow to split the orbiter apart. Both SRBs separated and flew uncontrollably until they were destroyed by the range safety officer.[2]

- NASA researched launch escape systems throughout development but determined that the Space Shuttle's predicted high dependability would eliminate the need for one. The two-person operators used modified SR-71 Blackbird ejection seats and complete pressure suits on the first four Space Shuttle orbital test flights, but they were eventually deactivated and removed for operational missions. Although practical flight escape systems were studied, they were rejected due to their complexity, high cost, and heavy weight. Following the disaster, a device was built to allow passengers to flee in glider flight, however this system would not have stopped an explosion during ascent.[1]

- The scheduled launch of the US space shuttle Challenger was deemed a "acceptable risk." The expression "acceptable risk" was used to the measure to be performed, not the choice to carry it out. If the concept of "acceptable risk" had been applied to the choice to launch, there could have been some ethical responsibility for the decision. The lack of moral behaviour in the launch choice was palpable.[1]

4. THE DAY BEFORE THE LAUNCH

The day before the Challenger launch, on January 28, 1986, NASA personnel were concerned about the weather forecast for the day of launch which was predicted to fall to about 20 degrees Fahrenheit and the ice formation on the launchpad. As a result, a teleconference was started at 8:45 p.m. EST was scheduled at Thiokol’s Utah facility with the Thiokol, Marshall, and Kennedy personnel. The teleconference focused on the issues of O ring behaviour at freezing temperatures. According to Thiokol's engineers, the low temperature would make the O-ring primary and secondary seals seal less fast, causing gases to leak out and lead to an

explosion on the launch pad once the SRBs ignited. But they failed in presenting the statistical analysis of the O-ring blow-by and the temperature in a manner convincing to managers since they lacked the data. The recommendation to not launch was vigorously defended by Roger Boisjoly and Arnie Thompson. The Thiokol Vice President Joe Kiliminister stated that he could not recommend the launch until it was 53 degrees Fahrenheit The data of the lowest temperature known was for the STC - 51C which was 53 F and thus the Thiokol engineers advised against launching until it reached at least 53 F while the expected launch temperature is way too less, 29 F and was way too unacceptable without the prior knowledge of the O ring stability. But the Marshall challenged Thiokol quoting the launch at 75 degrees Fahrenheit had a blow-by and hence making the point of independency of the temperatures with the o ring resiliency. The Thiokol manager Lawrence Mulloy questioned Thiokol making the statement “My God Thiokol, When do you want me to launch, Next April?”. Mr. Masons instructed Mr. Robert Lund, VP at Engineering Thiokol as “Take off your engineering hat and put on your management hat”. The statements listed above portray the stressful environment before the launch. Executives from NASA and Thiokol refused to delay the launch despite the data offered and the launch was then made to proceed without any delay. The launch signoff was first rejected by the Thiokol’s project manager, Allan McDonald, and another Utah official had to authorize on his behalf. One of the four corporate representatives, Mr. Robert Lund, 48, vice president for engineering at Thiokol, agreed to allow NASA clearance to launch Challenger despite the objections of staff engineers.[1]

5. CONSEQUENCES OF THE DISASTER

A presidential commission was held to examine the Challenger space shuttle STS 51-L which was appointed by President Reagan in February 1986 to deeply study and enquire about the possible causes of the explosion. This commission authored Rogers Commission after several months of investigation which was released on June 9, 1986. It urged NASA to focus on safety and prioritize the workforce and hence NASA created an office for safety, reliability, and quality assurance. NASA rebuilt the shuttle's solid rocket boosters to further prevent a repeat catastrophe of this magnitude. NASA was also urged to have proper communication between the engineers, managers, workforce, and all the institutions involved in it. A proper launch abort system and a crew escape system were made to be installed in it as a part of the crew safety measures. The process of retrieving Challenger's wreckage from the ocean and bringing it back to the Cape took 81 days. The Shuttle programme was put on hold for 32 months while various technological and management improvements were made to ensure the shuttle's safety. In 1991, NASA ordered the construction of the Endeavour to replace the Challenger. Recovery efforts after Challenger were estimated to have cost $12 billion.[1][3]

5.1 LEGAL IMPACT

The government and Morton Thiokol, the manufacturer of the solid rocket boosters, made compensation payments totalling $7,735,000 in cash and annuities to the families of the four astronauts who died in the Challenger catastrophe explosion. The Morton Thiokol and the government each contributed $4,641,000 and $3,094,000, respectively. Morton Thiokol's engineer, Roger Boisjoly, accused the company of several wrongdoings before a Presidential investigating commission and Congress, including fraud, carelessness, homicide, racketeering, defamatory comments, and untruthful testimony. Mr. Boisjoly sued Thiokol for $1 million in compensatory damages for his inability to work as an engineer and to compensate him for his mental health problems. A federal court has rejected two massive claims against the aerospace business brought by a former Morton Thiokol Inc. engineer in connection with the Challenger shuttle catastrophe. According to Winder, the federal judge "Boisjoly lacks standing to file a lawsuit and seek damages for the claimed misappropriation of public funds, misdirection of the space programme, losses incurred by insurance companies, harm to MTI stockholders, or astronaut deaths".[4][5]

5.2 ECONOMIC IMPACT

There are two main types of costs in a Challenger accident.
First are the reconstruction expenditures resulting from the inquiry, shuttle system improvements, and equipment replacement. Second are the operational expenses which are the expenditures and savings resulting from a lower-than-planned flying schedule. NASA has estimated that fixing the shuttle system will cost $240.5 million in 1986 and $245 million in 1987. These figures exclude the expense of additional orbiters. This is the first expense category, which includes expenditures for investigation, issue resolution (system adjustments), and equipment replacement. Most of these costs are related to changes to the shuttle system resulting from information collected as part of the survey. NASA predicts that it will cost at least $350 million. NASA projected that the probe would cost between $40 million and $65 million, with salvage expenditures, which were the most expensive part of the study, ranging from $30 million to $50 million. The cost of modifying the Shuttle Solid Rocket Boosters might be between $200 and $225 million. Estimated costs to procure new orbiter and replace the spare parts used to manufacture these orbiter projects cost NASA $263 million in 1986 and $702 million in 1987. A new budget authority was required, bringing the total cost to $2.4 billion.[6]

5.3 SOCIAL IMPACT

NASA's reputation suffered a massive blow, and it took years for them to gain back the governments and taxpayers trust back in the program. The live broadcast of the explosion provided a chance to assess whether or not normal latency-age children and adolescents would acquire symptoms of posttraumatic stress disorder (PTSD) and grieving following a distant, horrible occurrence. Moreover 60% of the individuals feared at least one Challenger-related stimuli within the first 5-7 weeks after the explosion. The East Coast and latency-age groups were far more symptomatic than the West Coast and adolescent groups. Most symptoms vanished during the course of the 14-month research. However, teenagers' lower expectations for the future in general grew, whereas latency-age children's modified attitude about space occupations remained rather stable. In 1986, three latency-age children from the East Coast satisfied the DSM-III-R symptom standards for PTSD.[7]

6. ETHICAL DILEMMA

The case study elaborates on the sequence of events that happened before the launch of the Challenger Space Shuttle. NASA was under enormous pressure from politicians and the other competing space agencies and due to this, the management was forced to make the launch despite the minimal testing and poor design of the O-rings. It is known that the launch was already postponed earlier for a couple of times which added extra pressure on management to make the launch. The Engineers at the Thikol company have warned the management about the faulty rings and about the unfavourable weather conditions but the management did not give an ear to the caution given which resulted in the explosion just after 73 secs after the launch of the Challenger Space shuttle claiming the lives of seven people and leaving a void in their families and a lifetime trauma to few people who tried to stop the launch. The crew members were un-informed about the faulty rings and the possible risks involved in the launch. The people in the higher management were kept in the dark about the meeting which had taken place between engineers and the managers of the Thiokol Company. And few other Ethical issues were Boisjoly took the documents home related to the disaster which can be considered as an act against the company's policy. No proper communication between the management and the engineers and an unhealthy management structure. Due to this accident, many people were affected and many have lost their trust in the NASA's reliability. The NASA and Morton Thiokol were held responsible for considering the business part and keeping the lives of the crew at risk which is definitely a violation of ethical values.[8]

7. WAYS THE ACCIDENT COULD HAVE BEEN AVOIDED

The engineers could have gone the extra mile to inform the higher management regarding the repercussions that would happen due to the faulty rings. The managers should have overcome the pressure and have heard and considered the arguments of the engineers which could have saved the lives of the astronauts. A proper escape plan should have been provided which could have helped the crew to land in safety during the explosion and the crew should have been warned about the possible risks involved due to the faulty rings. When the Presidential Committee questioned the astronaut's office member about the O-ring issue, the representative responded right away “We were not aware of any concern with the O-rings, let alone the effect of weather on the O-rings " which was not true. After months of investigation, it was brought to light that a phone call could have prevented the accident, but due to the negligence of a few people, this was never brought to the attention of the higher authorities. Bob Ebeling along with four other engineers insisted to postpone the launch for a couple of days until the weather is favourable but it was in vain, Bob had predicted that the space shuttle would blow off because the O-rings would not seal properly as a result of the cold weather he even told the same to his daughter on the day of the launch. It was the only launch that took place in such cold temperatures. Had the launch taken place a couple of weeks later it would have been such an achievement for NASA and as well the government. However, there was another possible course for the case to proceed. If the management had chosen more wisely, the explosion could not have happened. Although the manufacturer did not advise NASA to postpone the launching, they warned that the impact of launchpad ice on the shuttle was an unexpected occurrence that may threaten flight safety. But NASA persisted. For example, management might request extra testing and postpone its launching till the test is completed. It's probable that the launch was postponed so that design flaws in the shuttle could be addressed, so avoiding the accident. Their subordinates may have also had an effect on how the case progressed. Bringing the issue to the attention of upper management, for example, may have helped to resolve the communication breakdown. Thus, this may result in the launch being delayed or cancelled. Overall, the managers and staff should have prioritized safety by mandating further testing and design alterations. Stronger safety regulations and moral judgement would have assisted NASA in averting the disaster.[9]

8. ETHICAL FRAMEWORK

As an engineer, one must comply with myriad rules that safeguard the general public's safety and well-being. As a result, adhering to the current set of ethical norms is a must. Remember that engineers are responsible for the "safety, health, and well-being of the general public in the performance of their professional activities" since deciding on ethical norms that are most relevant to them. The mentioned code is critical to adhere to the established ethical standards since it encompasses the details of an engineer's responsibilities and functions, identifying certain of its components as potentially hazardous to bystanders. The general public's lack of

understanding about the harm that engineering tools and exposure to critical processes may pose renders the target population very exposed. As a result, as an engineer, one must protect those who are unaware of the dangers that engineering may bring to them from the aforementioned dangers. Another important criterion that engineers must follow to comply with their code of ethics is the scope of their duties. The decision to "provide services solely in their areas of expertise" and work only on problems directly connected to one's fields of competence is mature and critical to creating a successful disaster management plan and guaranteeing that none of the stakeholders involved experience negative repercussions. Furthermore, the defined rule was chosen as one of three key ethical criteria that an engineer must satisfy because of the influence it has on long-term results and an engineer's safety. The environmental effect of an engineer's performance is rarely considered important by the general public, but the consequences of a disaster caused by mishandling the relevant technical challenges can be disastrous. As a result, as an engineer, one must be aware of the possible consequences of one's performance on the environment. For example, an engineer must assess the likelihood of introducing garbage into the global environment as a potential hazard with potentially disastrous consequences. Because of electronic waste's negative impacts on the environment, adherence to the prescribed criteria is especially important in combating the possibility of a man-made disaster. As a result, whereas mismanagement of the two standards stated above appears to have a minor detrimental influence on public well-being, failing to address environmental problems has far-reaching consequences. The case's main ethical concerns were an absence of communication between management and poor safety protocols, which prevented sufficient technological testing before the launch. The primary ethical concern in this instance was a lack of an appropriate safety culture. Management's choice to launch the shuttle despite limited testing and defects in the O-ring design highlighted this. Management's decision breached the ethical rules of public welfare and human well-being since each of these variables increased the likelihood of an explosion during the launch. One of the most significant professional tasks of engineers is to adhere to these standards, as failure to do so may result in catastrophic tragedies [10][11].

9. STAKEHOLDERS

As stakeholders, we have the crew and NASA, who have a direct interest in the situation. Morton Thiokol and Rockwell are secondary stakeholders, as are any other indirect parties with an indirect interest in the matter. Managers and engineers at NASA and Morton Thiokol are also stakeholders because they were engaged in the design, communications, and decision-making processes leading up to the launch. Both sides suffered a considerable loss of confidence as a result of the occurrence. Finally, the government is a stakeholder since it has spent much on the program. Several technological issues combined to cause the Challenger space shuttle catastrophe. The primary technical problem was a failed O-ring seal upon take-off. when the SIR connection hardware and external fuel tank broke, pressurized, and hot gas leaked. Following that, the right-hand garment's aft attachment and the severe

damage to the fuel tank were separated. After then, the orbiter was easily destroyed by aerodynamic forces. Finally, the lack of an escape mechanism prevented the crew from ejecting from the shuttle after it was destroyed. The NASA authorities were aware of faults in the design of the Garb's O-rings for years but paid little attention to the matter. If the O-rings had been modified as soon as the defects were discovered, the technical issues might have been resolved and the accident avoided. In this situation, the broader public, additional staff members of the organization, and potential astronauts are secondary stakeholders. Because space shuttle initiatives like this one are paid for using public money, the general public is seen as a secondary stakeholder. Secondary stakeholders also include Morton Thiokol and other NASA workers who are connected to the businesses involved but are not actively participating in the project.[12][13]

10. DECISIONS AND ITS CONSEQUENCES

The government put ongoing pressure on NASA to have a low budget and highly predictable operations. Cost overruns and delays were evaluated more than mission accomplishment and safety. In the teleconference between NASA and Thiokol the day before the launch Thiokol's suggestion to push back the launch date was strongly rejected by NASA, and pressure was applied on them. Thiokol management approved the launch under duress from NASA, and Challenger was on the way to its fateful demise. The Challenger disaster's organizational cause was this breakdown in communication along with a management system that let NASA to bypass many safety regulations in order to speed up the launch . [14]

After months of investigation, it became clear that the calamity might have been avoided with just one phone call made that morning. It might have been given to Gene Thomas, the launch director, or Jesse Moore, the associate administrator for space flight at NASA. The explosion may have been avoided with better managerial judgement. For instance, Thiokol management may have vigorously opposed them, demanded additional testing, and delayed the launch until the required trials and testing are finished. To ensure that the mission's integrity would not be jeopardized, the organization ought to have undergone several dependability and quality assurance procedures before deciding to launch [1].

11.INVESTIGATION

The Challenger space shuttle and its crew were obliterated on January 28, 1986, in a blazing, catastrophic explosion, and NASA appointed members of the Rogers Commission find the root cause of the accident. Feynman accepted this commission's invitation to join it with some reluctance. He had no idea that he would be the one to identify the precise reason for the explosion. Theoretical physicist Richard Feynman is renowned for his contributions to particle physics, quantum electrodynamics, the theory of quantum electro fluidity, and the path integral formulation of quantum mechanics. In 1965, Feynman won the physics Nobel Prize in his contributions to the growth of quantum electrodynamics. He went directly to the

individuals who assembled the shuttle to inquire about what transpired. He picked up a lot of knowledge from these experts that would help him figure out what caused the explosion and information that made him aware of how dangerous flying a shuttle is. He considered it absurd for management to assert that there was a 1 in 105 chance of a disastrous failure of the shuttle. He had estimated the likelihood of a malfunction was roughly 1 in 100[4], which is a much different figure from what NASA has stated. He concentrated on the O rings, which were the main cause of the accident, in an effort to determine what went wrong. Feynman said that the NASA-specified Safety Factor of 3 for the O rings contained a brutal mistake. He eventually learned that when the temperature was below zero degrees, the O rings did not expand [4]. The accident occurred because the authorities did not take the weather on the launch day seriously enough. It was roughly 0 degrees that day, and the O rings failed as it lost its elastic properties under those conditions. To test his findings, he did the C clamp experiment in which he dropped the O ring material squeezed with a C clamp into a glass of cold water to simulate the actual conditions of the launch. With this he proved that the O rings cannot expand in 0 degrees thus leading to his conclusion. Also, out of 135 flights completed that he had observed till the end of his retirement, the shuttle experienced 2 catastrophic failures, or 1 in 67.5, which is not far from the prediction made before. In the end, he concluded that it was important to cease shuttle launches until NASA could overcome its internal discrepancies and provide an honest assessment of the shuttle's reliability based on his contacts with NASA management and engineers. This was brought about by the flagrant exaggeration of the risks, the major deficiencies in NASA management's grasp of science, and the lack of communication between the two groups. Feynman concluded with a comment that may succinctly summarize what must be done.[16]

“For a successful technology reality must take precedence over public relations, for nature cannot be fooled.” [15]

12. CONCLUSION

Using both the actor network theory of the STS framework and the ethics of responsibility framework, we can conclude that NASA managers, specifically Lawrence Malloy, the NASA project manager at SRB, are responsible for the explosion. This was demonstrated by identifying each actor in the network and checking whether the conditions of the accountability framework were met. A particular groupthink pattern that emerged from analysis of Challenger's decision was that the decision was made by NASA Level 3 administrators. They faced a provocative situation as it became increasingly difficult to maintain their flight plans. The decision-making group most clearly demonstrated her two manifestations of groupthink. Direct pressure on dissenters (Morton Thiokol's engineer) and mind guarding (O-ring concerns were not communicated to NASA Level 2 personnel). Finally, selective bias in processing existing information was the most obvious sign of erroneous decision making.[17]

The main lesson of this course is that ethical judgments must be postponed until the facts have been verified - and that hindsight and myths about ideal engineering practice must be avoided when determining which facts to verify and how to interpret them. We address numerous ethical considerations after detailing the events in the Challenger tale - both material and emotional issues:
1) Did NASA engineers breach their commitment to prioritize public safety over launch schedules?
2) Did the Thiokol engineers violate their commitment to prioritize public safety above fears about losing their NASA contract?
3) Has the concept of informed consent been violated?
4) What is the function of the whistleblower, and is there any bias towards whistleblowers?
5) Did Thiokol's engineer violate his commitment to treat papers as business property, pertaining specifically to his work?
6) What factors influence whether the public believes a technical judgment to be unethical? Answering these questions is difficult; this course attempted to demonstrate, using the Challenger case study, that the more you learn about the context of a technical disaster, the more difficult it becomes to accept single, simple explanations for what is happening and to assume that decision makers acted unethically.[18]

NASA has implemented a decision structure and related review process in the Space Shuttle program that appears to have all the necessary safeguards in place to prevent the risk from occurring in the case of Mission 51- L. This show's incredible streak of achievements reinforced the public's idea of an insecure system while concealing the fact that unwanted risks were taken almost from the start. These achievements appear to have contributed to NASA officials' willingness to continue taking risks, as well as the supposedly changing premise that missions must be proven unsafe rather than safe. Research initiated by the Challenger disaster has shown that longstanding problems have not been satisfactorily resolved and that even previously successful missions are most likely confused. The purpose of this investigation was to analyze the human factors that seem to have made the decision-making process ineffective, which, in theory, should lead to a different course of action. Our results show that no matter how carefully the decision structure appears in terms of the sequence of analysis and choices it appeals to decision makers, its effective use always depends on the social, psychological and communicative environment in which the responsible party operates. In the case of Challenger, there is significant evidence that the impact of these three areas played a role in the decision being made.
Several opportunities to avoid the launch had appeared in the previous 20 hours, but each time, one or more of the forces we identified came into play and reduced the chance of the collision trajectory changing. which NASA is fixing. A simple act of dissent by a potentially overconfident Level III official has undermined the respect NASA has earned and now must work to restore it.[14]

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14. SELF-REFLECTION

14.1. Megha Chandrashekharan

We have done a great job working together as a team. We first conducted a general investigation into the occurrence, spoke about our preliminary findings, and then split the topics among the group members. As a result, the job was evenly distributed, and each participant in the case study had a unique responsibility. We addressed the subjects at our often planned in-person and online sessions. As a result of careful planning, the report's completion went without a hitch.

14.2 Rini Susan Nutulapati

It's been a great experience working as a team and knowing in depth about the challenger space shuttle. Everyone had actively participated and made sure that the presentation and the report was done on time. We had weekly meetings and have distributed the work evenly, despite being from different domains we made sure that everyone was on the same page with respect to the topic. Every member worked earnestly and put in lots of effort for the completion of the report.

14.3 Abishek Ganesh

We have done a outstanding job of working on this case study and formed great companionship amongst ourselves along the way. During our research i have also come realise how lack of proper management could have a catastrophic effect and strive to do better. Lack of moral ethics and technical knowledge is a big problem everywhere and it has been reflected in this study as well. I have sworn to question any wrong decision in the company and not to come under peer pressure. Our case study report went smoothly and all of my teammates cooperated really well. We were able to coordinate our work as a result we were able to finish this report without any problems’

14.4 Vishwa Murugesan

We collaborated well as a group, and the team had frequent educational sessions. The team was able to allocate labour evenly based on each member's talents and past knowledge. The lack of in-person meetings was initially a hindrance, but once the online sessions started, the project was completed swiftly. Each member has significantly contributed time and effort to the report, and I believe they have done to the best of their ability despite any specific hurdles encountered along the way. This study helped me grasp how to do an ethical examination of an engineering problem.

14.5 Hariharan Shankar

We all got along well as a group, and the team often had productive meetings. Based on the individual abilities and prior work experience of each team member, the team was able to divide the workload fairly. Despite any specific roadblocks along the way, I feel that each member has contributed a great deal of time and effort to the report and has worked to the best of their ability. I learned how to analyse an engineering challenge responsibly after reading this report.

14.6 Shaivali Shinde

Overall, I believe the teamwork was excellent. We had weekly online and in-person team meetings where we could equitably distribute responsibilities and workloads among team members and also talk about how to effectively advance the project each week. Everyone put in a lot of time and attention into the report and made a big contribution. I learned how to analyse an engineering challenge responsibly after reading this report. The example report was a great help and gave us an idea about how to move forward with the case study. I learned how important ethical communication is between a scientific body.